

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

ABSTRACT:

This project develops an AI-based dynamic pricing system using Thompson Sampling. Different prices are treated as bandit arms, and the algorithm learns the best price from customer purchase behavior. It updates its decisions continuously to maximize revenue while adapting to changing market conditions.

INTRODUCTION:

Dynamic pricing adjusts product prices based on customer demand and market trends. Thompson Sampling helps select the most profitable price by balancing exploration and exploitation.

THEORY:

- Prices are treated as bandit arms.
- Customer purchase is the reward.
- Beta distributions are updated after each customer response.
- The algorithm gradually learns the optimal price.

OBJECTIVES:

- Build a dynamic pricing engine.
- Apply Thompson Sampling.
- Optimize prices using customer feedback.
- Increase revenue and conversion rate.

METHODOLOGY:

1. Define multiple price options.
2. Apply Thompson Sampling.
3. Observe customer purchases.
4. Update Beta distributions.
5. Repeat to improve pricing decisions.

DATASET:

- Product Price
- Customer ID

- Competitor Price
- Purchase (0/1)
- Revenue
- Time

THOMPSON SAMPLING MECHANICS:

- Initialize Beta distributions.
- Sample each price arm.
- Select the highest sampled price.
- Update the distribution based on purchase outcome.

COMPLEXITY ANALYSIS:

- Price Selection: $O(k)$
- Update: $O(1)$
- Overall: $O(k)$

RESULTS:

The model learns the best pricing strategy over time, improving revenue and customer conversion compared to fixed pricing.

APPLICATIONS:

- E-commerce
- Retail
- Airline Pricing
- Hotel Booking
- Ride-Sharing

CONCLUSION:

The proposed system uses Thompson Sampling to make intelligent pricing decisions. It adapts to customer behavior and market changes, providing efficient and profitable dynamic pricing.